

SALEM–SPENCER SETS AS A CROSS-PARADIGM SOLVER BENCHMARK: A WITNESS-SPLIT CP-SAT CAMPAIGN ON $r_3(212)$ AND A PARADIGM-RESISTANT RESIDUAL

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ABSTRACT. Deciding whether a 44-element subset of $[1, 212]$ can avoid all three-term arithmetic progressions is the current frontier instance of the Salem–Spencer sequence $r_3(N)$ (OEIS A003002) — a family of finite combinatorial feasibility problems that, unlike recent celebrated SAT-solved problems in Ramsey-type combinatorics, resists every off-the-shelf solver paradigm we tested. We present the problem family as a constraint-solving benchmark. Our campaign architecture combines a verified lower-bound witness, endpoint forcing, depth- d witness-variable splitting, window-cardinality inequalities derived from known $r_3(L)$ values, and recursive refinement of timed-out subproblems. Applied to $N = 212$, $K = 44$, it found no feasible 44-set across millions of CP-SAT subproblems, supporting but not proving the conjectured $r_3(212) = 43$. The window-cardinality family is the single most effective pruning layer, reducing the unresolved rate by 28 percentage points. Cross-paradigm analysis of the resistant residual shows a sharp stratification: an eight-hour HiGHS MIP audit closes 25 of 45 survivors while leaving 20 with dual bounds pinned at 0.0; pure CDCL closes 18 of those 20, each with an independently verified DRAT proof; and the final two subproblems (T1c) resist CP-SAT propagation, LP-based MIP, and CDCL under generous wall caps. We release tiered benchmark instances spanning this difficulty ladder, the verified DRAT/LRAT certificates, a deterministic instance generator, and a Lean formal-proof-search encoding of T1c, and frame the unit-gap problem $r_3(212) \in \{43, 44\}$ as a challenge for stronger inference, custom branch-and-bound, or formal proof search.

1. INTRODUCTION AND BACKGROUND

Let $r_3(N)$ denote the maximum size of a subset of $[1, N]$ containing no nontrivial three-term arithmetic progression. Equivalently,

$$r_3(N) = \max\{|A| : A \subseteq [1, N], \text{ no } a < b < c \text{ in } A \text{ satisfy } a + c = 2b\}.$$

The asymptotic study of $r_3(N)$ begins with Roth’s theorem [Rot53] and continues through Salem–Spencer and Behrend-type lower-bound constructions [SS42, Beh46] and modern density-increment upper bounds, including work of Bloom–Sisask and Kelley–Meka [BS20, KM23]. Those results describe the large- N behavior of progression-free sets. The present paper is about a different but complementary problem: exact finite computation at the OEIS A003002 frontier.

OEIS A003002 tabulates exact values of $r_3(N)$. Cariboni’s b-file currently reaches $N = 211$, where $r_3(211) = 43$ [OEI24, Car24]. The next natural decision problem is therefore:

Does there exist a 44-element 3-AP-free subset of $[1, 212]$?

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We found and independently verified a 43-element 3-AP-free subset of $[1, 212]$, so the lower bound $r_3(212) \geq 43$ is settled. The upper-bound question is whether $r_3(212) \leq 43$. Since $r_3(211) = 43$, any hypothetical 44-set in $[1, 212]$ must contain both endpoints 1 and 212; otherwise it would translate or restrict to a 44-set in $[1, 211]$. Thus the target is a single finite feasibility problem with a strong necessary endpoint condition.

Finite feasibility problems of this Ramsey-combinatorial flavor have been a showcase for modern SAT and constraint solving: the Boolean Pythagorean triples problem [HKM16], Schur number five [Heu18], and Keller’s conjecture in dimension seven [BHMN22] were each closed by massively parallel CDCL search with machine-checkable proofs. The Salem–Spencer family sits in the same problem class — Boolean variables, one short linear constraint per arithmetic progression, a cardinality target — yet, as this paper documents, its frontier instance behaves very differently: after the strongest problem-specific pruning we found, a small residual of subproblems resists CP-SAT propagation, LP-relaxation-based MIP, *and* CDCL under generous wall caps. That contrast is precisely what makes the family interesting as a benchmark: it is natural, scalable in N , has verifiable ground truth at solved sizes, and exposes a failure mode shared by all three mainstream complete-search paradigms.

We attacked this feasibility problem computationally using a reproducible CP-SAT architecture. The model uses Boolean variables x_i , one linear constraint for each 3-term arithmetic progression, the decision equality $\sum_i x_i = 44$, endpoint forcing, reflection symmetry breaking, and window-cardinality inequalities derived from the known values of $r_3(L)$ for $L \leq 211$. To make the search tractable, we split the problem by assigning high-degree variables from a verified 43-witness, generating a deterministic depth-24 chunk space, and then refine residual UNKNOWN chunks recursively.

The campaign did not produce a formal proof of $r_3(212) = 43$. It did produce three concrete outcomes.

First, we observed zero feasible 44-sets across millions of CP-SAT subproblems, including broad chunks, recursive refinements, wall-cap recalibrations, and targeted A/B experiments. This is empirical evidence for the expected value $r_3(212) = 43$, but it is not a certificate.

Second, we found that OEIS window-cardinality inequalities are essential. On controlled broad-pass ranges, adding these inequalities reduced the UNKNOWN rate by roughly 28 percentage points and also reduced aggregate solver time. They are the main successful pruning layer of the campaign.

Third, and most importantly, we identified a structural hard pocket. The largest retained broad run over 100,000 depth-24 chunks left 6,071 UNKNOWN chunks. A uniform random 100-chunk sample of those 6,071 was passed through a 300-second recap, leaving 45 resistant survivors. Re-attacking those 45 chunks with HiGHS, an LP-relaxation-based MIP solver, closed 0/45 in one-hour-per-chunk runs. A full eight-hour audit later closed 25/45, but left 20/45 UNKNOWN, all with dual bound still pinned at 0.0. A subsequent pure CDCL/SAT attack closed 18/20 of that LP-flat subset, while two chunks remained UNKNOWN even under a 12-hour pure-CDCL cap and a 4-hour windowed-CDCL diagnostic. Thus the obstruction is not merely a CP-SAT propagation artifact: the final audited pocket, T1c, isolates two chunks that resist every solver paradigm tested in this campaign.

This paper’s contribution is the benchmark and the cross-paradigm empirical study, not a new exact value of A003002. We provide:

1. A tiered benchmark release spanning a measured difficulty ladder: the 25 HiGHS-closable T1a chunks, the 18 CDCL-closable T1b-minus-T1c chunks, the 2 paradigm-resistant T1c chunks, the 6,071 broad UNKNOWN chunks from the 100k expansion batch, and a deterministic generator for the full depth-24 sweep.

2. A characterization of the structural hard pocket across three solver paradigms. The pocket initially appeared invariant under CP-SAT-side tuning and HiGHS; after the CDCL break, it remains invariant only on the final two-chunk residual T1c.
3. A reproducible witness-split plus window-cardinality architecture for exact $r_3(N)$ upper-bound search, parametric in (N, K) and reusable at adjacent frontier sizes. The problem-specific window-cardinality inequalities are, empirically, worth more than any generic solver lever we tested.
4. A detailed empirical account of the $N = 212$, $K = 44$ campaign, including broad-pass counts, refinement behavior, solver-time tradeoffs, and the controlled A/B experiments behind each design decision.

For reference, the benchmark tier notation used throughout the paper is:

Symbol	Meaning
T1	the 45 chunks that survived the 300-s recap of a random 100-chunk sample from the 6,071 broad UNKNOWNs
T1a	the 25 T1 chunks closed by the full 8-h HiGHS audit
T1b	the 20 T1 chunks left UNKNOWN by the full 8-h HiGHS audit, all with dual bound 0.0
T1b \ T1c	the 18 T1b chunks closed by CDCL and independently verified by DRAT checking
T1c	the final two chunks {40959, 48895} that resist all tested paradigms
T2	the 6,071 UNKNOWN chunks from the 100k broad expansion batch
T3	the full 12,582,912-chunk AP-pruned depth-24 sweep

1.1. Related work. Split-and-solve search. The witness-split architecture of §2.2 belongs to the cube-and-conquer family [HKWB12]: partition the search space by fixing a prefix of variable assignments (cubes), then solve each cube with a complete solver. The differences are the cube-selection policy and the refinement loop. Classical cube-and-conquer selects split variables by lookahead heuristics; we select them by AP-incidence degree *within a verified extremal witness*, exploiting the problem-specific intuition that a hypothetical 44-set must interact heavily with the densest regions of the known 43-set. Timed-out cubes are then re-split recursively (§2.4), which parallels iterative cube deepening.

Implied constraints and streamliners. The window-cardinality inequalities of §2.3 are implied constraints in the standard CP-modeling sense: valid inequalities not inferable by the solver from the base model. They are derived from previously *computed* values of the same sequence — the solved prefix of A003002 prunes the frontier instance. This is soundness-preserving, in contrast to streamlined constraint reasoning [GS04], which deliberately sacrifices completeness; it is closer in spirit to bound propagation from tabulated subproblem optima. Empirically these inequalities dominate every generic lever we tested (§3.2, §4.4).

SAT and CP in combinatorial mathematics. Beyond the celebrated results cited above, exact computation of Salem–Spencer values has its own lineage: Dybizbański [Dyb12] computed $r_3(N)$ through $N = 123$ with a tailored branch-and-bound, and Cariboni’s OEIS b-file [Car24] extends the sequence to $N = 211$. Our campaign differs in attacking the frontier with general-purpose solvers plus problem-specific pruning, and in reporting the negative space — which subproblems resist and why — rather than only the closed values.

Benchmarks and certified solving. Curated instance families — CSPLib [GW99], MIPLIB [GHG⁺21], the SAT Competition suites — have repeatedly steered solver research toward documented failure modes. The tiered release of §5 is offered in that tradition: a natural family, scalable in N , with verifiable ground truth at solved sizes and a measured cross-paradigm difficulty ladder. On the certification side, the CDCL-closed tier ships with DRAT proofs verified by `drat-trim` [HHB13, CFHH⁺17]; the absence of comparable certificates for the CP-SAT and MIP closures (§6.2) is an instance of the gap that auditable constraint solving [GMN22] aims to close.

1.2. Organization. The rest of the paper is organized as follows. §2 describes the architecture. §3 reports the computational campaign. §4 analyzes the hard pocket across HiGHS and CDCL. §5 formulates the remaining open problem as a benchmark release. §6 discusses transferability, limitations, and what the campaign suggests about finite r_3 upper-bound search.

2. ARCHITECTURE

The campaign rests on five components: a decision-form CP-SAT model of the 3-AP-free subset problem (§2.1), a witness-variable splitting scheme that produces tractable subproblems (§2.2), a window-cardinality pruning family derived from OEIS A003002 (§2.3), a recursive refinement loop for residual UNKNOWN chunks (§2.4), and the SLURM-side engineering that makes the workload reproducible (§2.5). Each component is independent of the others — any could be replaced by a stronger variant without disturbing the rest — and together they define the proof attempt on $r_3(212) \leq 43$.

2.1. Decision-form CP-SAT model. Fix $N \geq 3$ and $K \geq 3$. The standard CP-SAT encoding of “does there exist a 3-AP-free subset $A \subseteq [1, N]$ with $|A| \geq K$?” introduces Boolean variables $\mathbf{x_i} \in \{0, 1\}$ for $i \in [1, N]$ with $x_i = 1$ iff $i \in A$, and adds the constraints

$$\begin{aligned} \mathbf{x_a} + \mathbf{x_b} + \mathbf{x_c} &\leq 2 && \text{for every 3-AP triple } (a, b, c), \\ \text{sum_i } \mathbf{x_i} &\geq K. \end{aligned}$$

We adopt the **decision-form** variant in which the second inequality is replaced by the equality

$$\text{sum_i } \mathbf{x_i} = K.$$

The decision-form encoding is tighter for propagation, not for logical interpretation of UNKNOWN: an UNKNOWN return on either encoding is still consistent with both feasibility and infeasibility. The practical advantage of $\sum_i x_i = K$ is that CP-SAT and pseudo-Boolean propagators see both the lower and upper cardinality directions. Branches that would require more than K selected values, or too few remaining unfixed values to reach K , can be cut earlier than in the inequality form. For our purposes (proving the upper bound $r_3(N) \leq K - 1$), the equality form is therefore the right primitive.

For $N = 212$, $K = 44$, the model has 212 Boolean variables and 11,130 3-AP triple inequalities. We add the lex symmetry-breaking constraint

$$(\mathbf{x_1}, \mathbf{x_2}, \dots, \mathbf{x_N}) \geq_{\text{lex}} (\mathbf{x_N}, \mathbf{x_{\{N-1\}}}, \dots, \mathbf{x_1})$$

to factor out the reflection $i \mapsto N + 1 - i$. The opposite orientation would be equivalent; the implementation uses $\geq_{\text{lex}}$. We do not add other symmetries. After endpoint forcing, reflection is the only order-preserving symmetry we exploit.

We also apply **endpoint forcing** specific to the $r_3(212)$ case. Since OEIS A003002 records $r_3(211) = 43$, any 44-element 3-AP-free subset of $[1, 212]$ must contain both endpoints 1 and 212: if 1 were absent, the set would be a 44-element 3-AP-free subset of $[2, 212]$, which shifts to $[1, 211]$; if 212 were absent, it would already lie in $[1, 211]$. Either case contradicts $r_3(211) = 43$. We add $x_1 = x_{\{212\}} = 1$ as ground assumptions in every chunk.

Branching: variable selection by AP-incidence degree (the number of 3-AP triples a variable appears in), value selection `min`, and `fixed_search` to disable CP-SAT’s portfolio search. The first two target the most-constrained variables first; the third reduces solver-policy variation across runs, which is necessary for the A/B experiments of §3.2 and §3.5.

2.2. Witness-variable splitting. The decision-form model on the full range $[1, 212]$ is already at the limit of single-call CP-SAT solving in reasonable wall time. To break it into tractable subproblems we use a cube-and-conquer-style split [HKWB12] in which the cubes are chosen not by lookahead but from the verified 43-element lower-bound witness $A_{43} \subset [1, 212]$ (§2.1, §3.1). The splitting proceeds in three steps.

Degree ranking. For each $v \in A_{43}$, compute the AP-incidence degree $\deg(v) = |\{(a, b, c) : v \in \{a, b, c\}, b - a = c - b\}|$. Sort A_{43} by $\deg(v)$ in decreasing order. Pick the top $d = 24$ witness values; we refer to this prefix as the **broad split prefix**.

Combinatorial enumeration. For each of the $2^{24} = 16,777,216$ IN/OUT assignments of the broad split prefix, instantiate a **chunk**: the decision-form model with x_v fixed to the chosen value for every v in the prefix. A chunk is identified by its 24-bit chunk ID.

AP-prefix pruning. Many chunks are immediately infeasible at the 3-AP level alone: if the broad split prefix forces three pinned-IN values that form a 3-AP, the chunk has no feasible completion and need not be sent to CP-SAT. We perform this check during chunk emission and skip such chunks. The surviving chunk count for $N = 212$, $K = 44$ at depth 24 is 12,582,912, roughly 75% of the raw 2^{24} count.

The choice $d = 24$ is empirical: larger d produces more chunks but each is easier; smaller d produces fewer chunks but each is harder. At $d = 24$ the per-chunk solver time at the 60-s wall cap has a heavy tail but a tractable median (the broad pass closes ~94% of expansion chunks within 60 s — see §3.1), which makes the depth 24 choice a workable broad layer.

2.3. Window-cardinality pruning from OEIS A003002. The 3-AP-free subset problem admits a family of implied constraints — valid inequalities that are not derivable by the solver from the 3-AP triple inequalities alone. For any window $[a, a + L - 1] \subseteq [1, N]$ and any length L with $r_3(L) < L$,

$$\sum_{i \in [a, a+L-1]} x_i \leq r_3(L).$$

This is valid because $\{i \in A : i \in [a, a + L - 1]\}$ is itself a 3-AP-free subset of an L -element interval, hence of size at most $r_3(L)$. Crucially, the right-hand side $r_3(L)$ is *constant* with respect to a , so the family is a single constant per window length, not a learned bound.

We populate the right-hand sides from the OEIS A003002 b-file (Cariboni’s tabulation, valid for $L \leq 211$). For $N = 212$, $K = 44$ we generate window constraints for every available length L with $r_3(L) < \min(L, K)$ and every $a \in \{1, 2, \dots, 212 - L + 1\}$. The resulting model contains 22,154 window inequalities.

Empirically the window family is the single most impactful CP-SAT intervention in the campaign: the controlled A/Bs of §3.2 show a ~28-percentage-point reduction in UNK rate at the 60-s wall cap when window-bounds are enabled. We use it as the default configuration for every solver call beyond the calibration batch.

2.4. Recursive refinement. A chunk that returns UNKNOWN at the broad layer is refined by extending the broad split prefix with the next m witness values (by AP-incidence degree) and re-emitting the resulting 2^m subchunks (less AP-prefix pruning) to the solver under a longer wall cap. The depths and per-step parameters used in the campaign are:

Level	Depth	Step size m	Per-chunk wall cap
Broad	24	—	60 s
L1	40	+16	60 s
L2 (tail)	48	+8	60 s
L3 (level3)	56	+8	60 s
L4 (level4)	64	+8	600 s

Each level is fed only the UNKNOWN residuals of the prior level, so the work at deeper levels is concentrated on the hard tail. The expected per-level fan-out 2^m is mitigated by AP-prefix pruning, which becomes more aggressive at deeper levels (more pinned-IN values, more chances for a 3-AP among them). For example, the Sample-500 L1 stream emits 2,076,105 rows from 500×2^{16} nominal descendants, a ~6% survival rate after pruning.

Refinement is the campaign’s primary tool for closing residual UNKNOWN chunks. It reliably closes individual deep chunks (§3.1 reports 6/6 closure at L4). It does not generalize cheaply to the full 6,071-chunk expansion residual because the per-chunk cost grows roughly geometrically with depth.

2.5. SLURM engineering. The campaign runs on the UMass Unity SLURM cluster. The workload pipeline is implemented as a small collection of Python scripts, each with a single responsibility:

- **r3_slurm_emit.py** — generates the chunk-ID list for a given (N, K, split-vars, depth, range) and emits a SLURM array sbatch script that fans out the chunks across array tasks. The `--chunks-per-task` flag controls the granularity of fan-out.
- **r3_split_cpsat.py** — the per-task driver. Reads its slice of chunk IDs, instantiates the CP-SAT model for each, runs the solver under the per-chunk wall cap, and appends one JSONL row per chunk to a shard file.
- **r3_collect.py** — merges shard files into the canonical batch output. Handles deduplication if a chunk was solved more than once (e.g., during retries).
- **r3_tail_emit.py** — emits the next-level refinement workload from the UNKNOWN rows of a parent batch. Supports multi-parent inputs (each input JSONL contributing its own parent prefix) and templated output naming via a parent-tag regex.
- **r3_proof_manager.py** — tracks the proof-state graph across levels, identifying which chunks have been closed at which depth and which remain open.
- **r3_verify.py** — independent triple-enumeration verifier for any candidate K -element witness.

Two engineering decisions are worth highlighting because they affect reproducibility:

Atomic shard renames. Each per-task driver writes its shard to a temporary `.tmp` path keyed by the SLURM job and array IDs, and renames it to `shard.jsonl` on successful completion. A killed or pre-empted task leaves only the temporary file, which is ignored by the collector. This pattern prevents partial output from corrupting downstream collection without requiring any locking.

Deterministic inputs. Every chunk ID maps to a fixed prefix assignment, and every CP-SAT call uses the same solver seed and fixed-search branching. This is necessary for the lever A/B experiments of §3.5: a “no measurable effect” claim is only defensible if the baseline arm is reproducible under the same software stack.

The end-to-end pipeline is driven by `unity_handoff.sh`, which runs the broad pass, the recap studies, the refinement levels, and the HiGHS attack as separately resumable phases. A complete campaign reproduction on a fresh allocation requires the OEIS b-file, the verified 43-witness, the

Python environment (OR-Tools, `highspy`, PySAT/CaDiCaL, and `drat-trim`), and a SLURM allocation; everything else is generated by the scripts.

3. EMPIRICAL CAMPAIGN

This section reports the workload, results, and key A/B tests of the $r_3(212) \leq 43$ campaign. All experiments were run on the UMass Unity SLURM cluster against the architecture described in §2. The headline numbers: millions of CP-SAT subproblems solved, 0 **FEASIBLE** rows, and a 6.07% **UNKNOWN** residual in the largest retained broad batch that motivates the structural analysis in §4.

3.1. Monolithic baseline. Before any splitting, we hand the full unsplit decision instance — 212 Boolean variables, 11,130 3-AP inequalities, $\sum_i x_i = 44$, endpoint forcing, with and without the 22,154 window-cardinality inequalities — to each solver paradigm with a 24-hour wall cap.

Paradigm	Windows	Wall	Status	Progress at cap
CP-SAT (8 workers)	on	24 h	TODO	TODO conflicts
CP-SAT (8 workers)	off	24 h	TODO	TODO conflicts
HiGHS MIP (8 threads)	on	24 h	TODO	TODO nodes, dual TODO
HiGHS MIP (8 threads)	off	24 h	TODO	TODO nodes, dual TODO
CDCL (1 core)	on	24 h	TODO	TODO conflicts
CDCL (1 core)	off	24 h	TODO	TODO conflicts

No paradigm resolves the monolithic instance within the cap, which is the empirical justification for the witness-split architecture of §2.2: the campaign’s progress comes from the split and the window bounds, not from raw solver time on the whole problem.

3.2. Chunk-count breakdown. The campaign proceeded in three layers of increasing scope, plus a recursive-refinement loop on the **UNKNOWN** residuals.

Verified lower bound. A 43-element 3-AP-free subset of $[1, 212]$ is recorded in the repository’s witness JSON and re-verified by `r3_verify.py` against an independent triple-enumeration check. This witness fixes the campaign’s $K = 44$ decision-mode target and supplies the seed for the depth-24 broad split (§2.2).

Broad pass at depth 24, 60-s wall. The retained broad-pass logs include both pre-window and window-bound runs:

Range	Chunks	INFEASIBLE	UNKNOWN	UNK rate
Calibration, no window bounds, [575, 1575)	1,000	799	201	20.10%
Bounded, no window bounds, [1575, 11575)	10,000	6,967	3,033	30.33%
Bounded, with window bounds, [1575, 11575)	10,000	9,835	165	1.65%
Expansion, with window bounds, [11575, 111575)	100,000	93,929	6,071	6.07%

The 20.10% calibration rate was run *before* window-bounds were enabled and is included for reference only; once window-cardinality pruning is on (see §3.2), the residual sits in the 1–6% band depending on the chunk-ID range.

Recursive refinement of UNKNOWN chunks. Every chunk timing out at the broad layer can be refined by extending the witness-pin prefix with the next-degree split variables and re-solving the resulting subchunks. The number of emitted rows is not a raw 2^d fan-out: AP-prefix pruning eliminates many descendants before a solver call is made. The retained refinement diagnostics are shown below. The depth labels follow the L1/L2/L3/L4 refinement ladder of §2.4.

Stream	Source	Depth added	Rows emitted	INFEASIBLE	UNKNOWN
Sample-100 L1	random sample of 100 broad UNKs	+16 (to depth 40)	299,375	299,374	1
Sample-100 tail	one Sample-100 residual	+8 (to depth 48)	8	8	0
Sample-500 L1	stratified sample of 500 broad UNKs	+16 (to depth 40)	2,076,105	2,076,095	10
Sample-500 tail8	10 Sample-500 residuals	+8 (to depth 48)	76	68	8
Sample-500 level3	8 tail8 residuals	+8 (to depth 56)	8	2	6
Sample-500 level4	6 level3 residuals at 600-s wall	+8 (to depth 64)	6	6	0

The final Sample-500 level4 cleanup closed all 6 deep residuals within the 600-s cap, with a worst-case solver time of 599.74 s — within 0.26 s of the cap, so this cleanup is at the practical limit of the refinement strategy at its current parameter settings. Crucially, none of these levels produced a **FEASIBLE** row.

The 6,071 UNKNOWN chunks of the expansion batch were *not* fully refined by this campaign. Three subsets of them were sampled for diagnostic experiments: a uniform random 100-chunk recap study (§3.2 and §4.2), a structural-mining analysis (§4.1), and the HiGHS attack of §4.3.

3.3. The window-cardinality A/B. The single most impactful intervention in the campaign was adding window-cardinality inequalities derived from the OEIS A003002 b-file. For every window $[a, a + L - 1] \subseteq [1, 212]$ and every length L with $r_3(L) < \min(L, K)$, we add $\sum_{i \in \text{window}} x_i \leq r_3(L)$. For $N = 212$, $K = 44$ this generates 22,154 inequalities, on top of the 11,130 3-AP triple inequalities and the symmetry-breaking lex constraint.

We measured the effect on three batches of varying size:

Batch	Chunks	UNK without window-bounds	UNK with window-bounds	Reduction
[575, 675)	100	27.00%	0.00%	-27.0 pp
[1575, 11575)	10,000	30.33%	1.65%	-28.7 pp
[11575, 111575)	100,000	n/a	6.07%	—

Both controlled A/Bs show a roughly 28-percentage-point reduction in UNK rate at the 60-s broad wall cap. The window-bound family is strictly stronger than the baseline 3-AP family at this problem size. In the retained logs it also lowered total broad-pass solver time, because many formerly hard chunks closed early under the extra bounds.

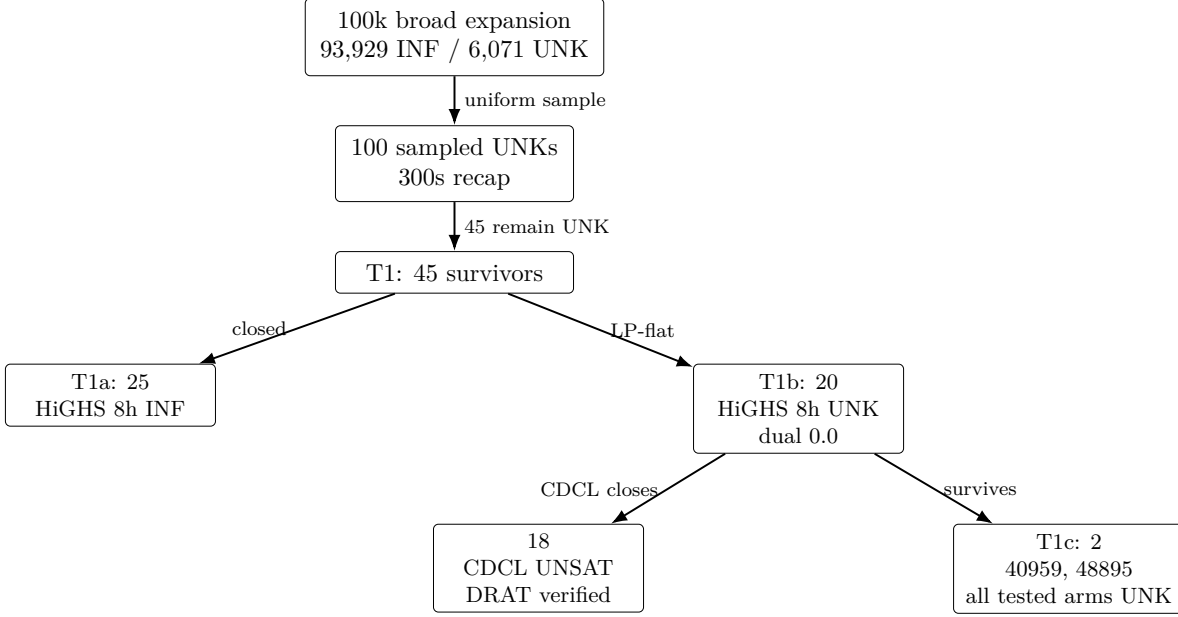


FIGURE 1. Funnel from the retained ‘100k’ broad expansion batch to the final two-chunk T1c residual. T1 is a ‘45’-chunk subset produced by recapping a uniform random ‘100’-chunk sample of the ‘6,071’ broad UNKNOWN chunks at ‘300’ seconds.

The $[575, 675)$ A/B is a 100-chunk subrange of the calibration range reported in §3.1; the difference between its 27.00% no-window UNK rate and the calibration range’s 20.10% reflects the structural non-uniformity of chunk-ID space rather than a change in model configuration.

However, the UNK rate is non-monotone in the chunk-ID range: the small controlled A/B on $[575, 675)$ closes to 0%, the 10k bounded batch sits at 1.65%, and the 100k expansion batch sits at 6.07%. This indicates that the chunk-ID space is structurally non-uniform — later chunk IDs (which encode different witness-pin assignments) are harder in a way that window-bounds do not fully compensate for. The bucket analysis in §4.1 quantifies this non-uniformity.

3.4. Solver configuration and engineering. All CP-SAT calls use the following configuration unless explicitly overridden in an experiment:

- Model: decision-mode $\sum_i x_i = 44$, 11,130 3-AP linear inequalities, 22,154 window-cardinality inequalities, lex reflection symmetry break, endpoint forcing $x_1 = x_{\{212\}} = 1$.
- Branch strategy: variable selection by AP-incidence degree, value selection `min`, `fixed_search` to disable CP-SAT’s portfolio.
- Solver: OR-Tools CP-SAT [PF24], 8 workers per task, fixed solver seed and fixed search parameters, wall cap as noted (most commonly 60-s for broad, 300-s and 600-s for recap experiments).

The SLURM emitter, shard collector, tail emitter, and proof-state manager implement the workload pipeline. Output is written as line-delimited JSON with one record per chunk; shards are written to a temporary path and atomically renamed on completion so that partial output from killed tasks cannot corrupt downstream collection. The full campaign is reproducible end-to-end via the `unity_handoff.sh` driver.

3.5. Zero FEASIBLE across the campaign. Aggregating across the broad pass, the refinement loop, the recap studies, the lever experiments of §3.5, and the HiGHS attack of §4.3, the campaign solved millions of $(N, K, \text{fixed_in}, \text{fixed_out})$ CP-SAT subproblems, including prefix-closure

work not tabulated above, 45 HiGHS MIP subproblems, and the CDCL/SAT and proof-producing reruns reported in §4.3. The number of subproblems returning **FEASIBLE** is **zero**.

The **FEASIBLE** count is the only signal that would directly disprove $r_3(212) \leq 43$; its absence does not constitute a proof of the upper bound, but it is strong empirical support, especially given that the campaign forced the endpoints 1 and 212, which any 44-set must contain by the known value $r_3(211) = 43$, and then explored a depth-24 prefix of the verified 43-witness in the processed chunk ranges. Within the processed expansion range, a 44-element 3-AP-free subset of $[1, 212]$, if one existed, would have to lie in the 6,071 unresolved broad **UNKNOWN** chunks (or in their deeper refinements). Globally, the much larger unprocessed remainder of the full depth-24 sweep remains open as well.

3.6. Search-tuning levers and their effect sizes. We tested five CP-SAT-side search-tuning levers. The full lever inventory, including the HiGHS substitution and the pair-AND Tseitin experiment, is consolidated in §4.4; here we report only the three CP-SAT-side levers that operate at the broad layer.

Lever	Mechanism	Bucket	Baseline UNK	Treatment UNK
Split-vars reorder	Permute the depth-24 prefix so hot pins (values {68, 75, 70, 76, 91}) lead	[61575, 66575)	13.60%	12.96%
Wall-cap extension 60s → 300s	Same model, same prefix, longer per-chunk budget	random 100 from expansion UNK	100.00%	45.00%
Recursive deepening	Refine UNK to depth 64 at 600-s cap	6 L3 residuals	100.00%	0.00%

The reorder lever was tested for completeness but is expected to be a null result on principle: the depth-24 broad split iterates over the 2^{24} IN/OUT prefix assignments, and within-prefix variable ordering does not change the set of surviving chunks under AP-prefix pruning — only the order in which they are emitted to SLURM. The within-noise result confirms this.

The wall-cap extension lever moves the residual substantially on the first doubling but exhibits a saturating closure curve under further extension. The 60s → 120s → 300s series is analyzed in §4.2; in brief, the marginal close-rate per additional second of wall drops sharply past 300-s, which is why the campaign did not pursue 600-s or 1200-s broad-layer reruns on the full 6,071-chunk expansion residual.

The recursive-deepening lever, applied locally to small residual sets, reliably closes individual deep UNK chunks (6/6 at the L4 level in §3.1). But the levers it implies — picking the next-degree split variables and re-solving up to 2^8 descendant assignments at a longer wall — do not generalize cheaply to the full 6,071-chunk expansion residual. A naïve depth-32 rerun on every expansion UNK would emit roughly 1.5 million subproblems and require an order of magnitude more CPU time than the original broad pass, with no guarantee of closing the hard core analyzed in §4.

3.7. Compute budget. The dominant retained costs are the 100k broad expansion, the sample-500 refinement, and the full-T1 long-wall HiGHS audit; the CDCL and certificate diagnostics are comparatively cheap. The total retained-log estimate is roughly 7,850 worker-hours; Appendix A itemizes the accounting per campaign layer. A full depth-24 sweep of the 12,582,912 AP-pruned chunks of the witness-split lattice would require orders of magnitude more; more importantly, it

is not justified by the present evidence: the hard pocket of §4 is the binding obstacle, not simply broad-layer throughput.

4. THE STRUCTURAL HARD POCKET

The headline empirical finding of the campaign is not the absence of a 44-element 3-AP-free subset of $[1, 212]$ — which we expected from the OEIS A003002 extrapolation $r_3(212) = 43$ — but the existence of a small, robust subset of broad subproblems that resist every solver paradigm and search lever we threw at them. We refer to this subset as the *hard pocket*. This section characterizes it empirically, summarizes the unsuccessful interventions, and states the resulting open problem.

4.1. Empirical signature on the broad population. The 100,000-chunk window-bound broad batch over chunk-id range $[11575, 111575)$ produced 93,929 **INFEASIBLE** chunks, 6,071 **UNKNOWN** chunks, and 0 **FEASIBLE** chunks under a 60-second per-chunk wall cap. The 6.07% **UNKNOWN** rate is non-uniform in two distinct senses.

Bucket non-uniformity. Partitioned into twenty 5,000-chunk buckets, the **UNKNOWN** rate varies from 1.64% to 13.60%, with the worst bucket $[61575, 66575)$ at 13.60%. The longest contiguous **UNKNOWN** run found in this batch was length 27 at chunk IDs 98277..98303; many other runs have length around 15. These contiguous runs are a strong indication that **UNKNOWN** status is driven by joint structural properties of the depth-24 witness-pin assignment, not by stochastic solver behavior.

Pin-OUT enrichment. For each of the 24 broad-split witness variables we computed the conditional **UNKNOWN** rate given that variable’s IN/OUT assignment in the chunk. Five variables — values 68, 75, 70, 76, 91, all in the dense middle cluster $[67, 91]$ of the verified 43-witness — show a strong asymmetry: their conditional **UNKNOWN** rate is roughly 2.4% when pinned IN and 9.7% when pinned OUT (Fisher log-odds +3.5). The signal is consistent across the five values. We refer to a chunk in which all five hot pins are forced OUT as a *middle-out chunk*. Middle-out chunks account for a disproportionate share of the **UNKNOWN** tail of the broad batch.

4.2. The 300s-resistant tail does not collapse. To test whether the broad-pass **UNKNOWN**s were merely “slow” rather than structurally hard, we ran two wall-cap recalibration experiments on a uniform random sample of 100 **UNKNOWN** chunks drawn from the 6,071 baseline. The results show a saturating, not exponential, closure curve:

Wall cap	INFEASIBLE	UNKNOWN	Close rate over baseline
60 s	0	100	0%
120 s	31	69	31%
300 s	55	45	55%

Going from 60s to 120s (2x) closed 31 chunks. Going from 120s to 300s (2.5x further) closed only 24 more. Naïve extrapolation to 600s predicts perhaps 10–15 additional closures. The hard tail does not vanish under more time.

We applied the same structural mining procedure (single-pin, pair, and triple log-odds enrichment relative to a matched **INFEASIBLE** sample) to the 45 **UNKNOWN** chunks that survived the 300s cap. The broad pin-OUT signature weakens substantially. The best high-coverage pair, $[91 = \text{OUT}, 48 = \text{OUT}]$, covers 66.67% of the survivors with log-odds +1.226 — a modest effect size, far from the +3.5 signal seen on the 6,071-row population. Top triples have stronger log-odds but each covers only a handful of cases. Hamming-distance clustering of survivor **fixed_in** sets shows multiple small clusters rather than a single dominant niche. We interpret this as evidence that the 300s-resistant

subproblems share moderate pin-OUT structure but spread across many distinct sub-pockets of the depth-24 assignment lattice.

4.3. LP-paradigm failure and the CDCL break. The 300s-resistant pocket might in principle be a CP-SAT-specific phenomenon — an artifact of constraint propagation’s inability to exploit implicit LP structure. To test this, we re-attacked all 45 survivors with the HiGHS open-source MIP solver [HH18], which combines branch-and-bound, LP relaxation, presolve, cutting planes, and primal heuristics. We used a one-hour wall cap per chunk, 8 threads per chunk, and the identical constraint set (decision-form $\sum x_i = 44$, all 11,130 3-AP triple inequalities, all 22,154 window-cardinality inequalities, plus the broad chunk’s `fixed_in` and `fixed_out` assignments tightened into variable bounds).

The one-hour result was negative:

The two rows in the following table are not a common-sample comparison: the HiGHS attack targets precisely the 45 chunks that survived the 300-s CP-SAT recap, i.e. the harder subset of the 100 recap inputs.

Solver	Constraints	Wall cap	Closed (INF)	UNKNOWN	Aggregate solver wall time
CP-SAT, window- bounds	3-AP + window + symmetry	300 s	55 / 100	45 / 100	bounded by ~8.3 wall-h
HiGHS, window- bounds	3-AP + window + endpoint	3,600 s	0 / 45	45 / 45	45.0 wall-h

Across 162,003 solver-seconds and 3,181,316 explored MIP nodes, HiGHS did not close a single chunk. No **FEASIBLE** row appeared. The recorded HiGHS dual bound stayed at its uninformative default value in every task, which is itself evidence that the LP-relaxation route was not discovering a useful certificate for these instances.

We then re-attacked all 45 T1 chunks under an extended 8-hour HiGHS wall, with LP progress logging enabled. The result is mixed: 25/45 chunks closed **INFEASIBLE**, while 20/45 returned **UNKNOWN** at the full cap with dual bound still pinned at 0.0. We call this 20-chunk LP-paradigm-resistant subset **T1b**. The audit consumed 901,073 solver-seconds and 25,196,448 MIP nodes.

To test whether T1b is invariant under solver architecture more broadly, we re-attacked it with a CDCL/SAT solver (CaDiCaL via PySAT [BFFH20, IMMS18], single-threaded, 4-hour wall, encoding restricted to 3-AP triples + cardinality + chunk pins; no window-cardinality clauses). CDCL closed 18/20 chunks **UNSAT**. We call the surviving 2-chunk residual **T1c** = {40959, 48895}. A T1c diagnostic at extended wall (12 h pure CDCL) and with totalizer-encoded window constraints for lengths {31, 100, 199} (4 h) also returned **UNKNOWN** on both chunks, so T1c is resistant to all tested paradigms in this campaign.

A proof-producing rerun of the 18 CDCL-UNSAT chunks initially emitted DRAT proofs for 17 chunks; the remaining chunk, 32735, required a larger-memory proof rerun and then also emitted a DRAT proof. Independent `drat-trim` verification confirmed all 18/18 emitted proofs. The final two certificates were the slowest: 63231 verified in 49,758.11 seconds and 32735 verified in 80,960.68 seconds under the final 24-hour verifier pass [HHB13, CFHH⁺17].

The refined paradigm-invariance picture is therefore: **LP-paradigm methods (CP-SAT constraint propagation and HiGHS LP-relaxation MIP) fail uniformly on T1b**, and the CDCL clause-learning paradigm closes 18/20 of those chunks, all of which are independently verified. The genuinely paradigm-invariant residual is T1c, of size 2.

4.4. Levers tested and their outcomes. For completeness, we record every search-tuning lever we tested on the hard bucket [61575, 66575) or the 300s-resistant subset. The CP-SAT/MIP-side tuning levers did not remove the hard pocket; the one qualitative exception is the CDCL paradigm switch, which closes most of T1b and therefore narrows, rather than merely tunes, the residual.

Lever	Mechanism	Result
Window-cardinality (OEIS A003002)	Add $\sum x \leq r_3(L)$ for each window	UNK rate on full 10k batch: 30.33% $\rightarrow$ 1.65%. Strong but plateaus.
Split-vars reorder (hot pins first)	Permute the depth-24 split prefix	13.60% $\rightarrow$ 12.96% on the worst bucket. Noise.
Wall-cap extension	60s $\rightarrow$ 300s on the broad pass	Saturating curve, see §4.2.
Targeted pair-AND	Explicit <code>pair[a,c] =</code>	Control 67.84s, treatment 71.75s on a
Tseitin propagators	<code>x[a] ^ x[c]</code> BoolVars on midpoints in [67, 91]	sampled residual. No effect.
Recursive deepening	Refinement at depths 40, 48, 56	All 500 stratified-sample base chunks close, but with a non-trivial tail; the final six level-3 residuals needed up to 599.74s at the 600s cap.
HiGHS MIP with LP/cut machinery	Replace CP-SAT entirely	0/45 closed at 1 hour; 25/45 closed in the full 8-hour audit, while 20/45 retained dual bound 0.0.
Pure CDCL/SAT	Encode T1b as CNF with 3-AP clauses + cardinality + pins, no windows	18/20 HiGHS-flat chunks closed UNSAT in 4 hours; 2/20 remained UNKNOWN; no SAT rows. This breaks the strong solver-invariance framing.

The pair-AND result is worth a closer note. The added constraints are mathematically redundant with the existing 3-AP triple inequalities — they encode the same forbidden configurations, just with an explicit Tseitin variable per pair. The hope was that this would give CP-SAT more “propagation hooks” in the structural pocket. The negative result is consistent with the broader picture: the pocket’s hardness is not an encoding issue, it is a search-space issue.

4.5. A conjecture about the pocket. We end §4 with a working conjecture about T1c.

CDCL closed 18/20 of T1b, refuting the strongest reading of solver-paradigm invariance for the full 20-chunk LP-flat subset. The conjecture below is deliberately narrower: it concerns only the two chunks that survived the CDCL break.

Each T1c subproblem corresponds to a depth-24 fixed assignment in which (i) the LP relaxation upper bound on $\sum x_i$ is at most one above the decision threshold $K = 44$, leaving no room for LP-based cuts to prove infeasibility, and (ii) the integer infeasibility certificate is too combinatorially diffuse to fit in the working memory of current CDCL clause-learning solvers under a 12-hour wall. The audit of §4.3 bounds the T1c population at 2 chunks within the 100,000-chunk window-bound expansion residual; we do not have a population-level estimate beyond the audited region.

If this picture is correct, closing T1c requires either (a) a problem-specific additive-combinatorial upper bound tighter than the OEIS window-cardinality family, or (b) a custom branch-and-bound or proof-search system that exploits problem structure neither current general-purpose solver paradigm captures. Both are framed as open computational problems in §5.

4.6. What this means for $r_3(212)$. The combined CP-SAT, HiGHS, and CDCL evidence — 0 FEASIBLE/SAT rows across millions of subproblems and all audited hard-pocket diagnostics — strongly supports the OEIS-conjectured value $r_3(212) = 43$, but the surviving T1c pocket means we do not have a formal proof of the upper bound $r_3(212) \leq 43$. The 6,071 baseline UNKNOWN chunks (and the much larger unobserved tail of the full 12,582,912-chunk depth-24 sweep) remain an open obstruction. The lower bound $r_3(212) \geq 43$ from the verified witness in §2 is unaffected. The trivial upper bound $r_3(212) \leq 44$ follows from monotonicity and OEIS $r_3(211) = 43$, so the gap between our certified bounds is exactly one.

Closing that final unit gap would require resolving T1c and independently certifying the entire depth-24 sweep, including the unprocessed remainder of the 12,582,912-chunk T3 lattice. T1c is the sharpest audited open instance, not necessarily a sufficient stepping stone. We release the hard-pocket benchmark instances in §5 so specialists can attempt this directly.

5. OPEN PROBLEM AND BENCHMARK RELEASE

The campaign establishes strong empirical support for $r_3(212) = 43$ (§3.4) but leaves a hard pocket that resists both CP-SAT and HiGHS under every lever we tested (§4). To turn that pocket into a tractable research target rather than an open compute frontier, this section states the residual problem cleanly and releases the surviving instances as a tiered benchmark.

5.1. Statement of the open problem.

Problem (R3-212-UB). Let $A \subseteq [1, 212]$ with $|A| = 44$. Determine whether there exists A containing no nontrivial 3-term arithmetic progression. The verified 43-element witness in `results/N212_K43_witness.json` (§2.1) and the value $r_3(211) = 43$ from OEIS A003002 jointly imply that any such A must contain both endpoints 1 and 212. Equivalently, either exhibit one such A or certify the infeasibility of the decision problem $\sum_i x_i = 44$ over the 3-AP constraint family on $[1, 212]$ with $x_1 = x_{\{212\}} = 1$.

A solution to R3-212-UB resolves the unit gap $r_3(212) \in \{43, 44\}$ in either direction. Our multi-million-subproblem CP-SAT campaign and the 45-instance HiGHS attack returned 0 FEASIBLE rows, which is evidence for the infeasibility branch but not a proof. The benchmark instances released below pinpoint the subproblems on which generic combinatorial-optimization solvers currently fail.

5.2. Benchmark instance tiers. We release three tiered benchmark sets. Each JSONL row records the chunk ID, fixed assignments, witness-pin prefix, solver status, and timing data; the common constraint family (3-AP triples, endpoint forcing, and OEIS window bounds) is reconstructed by the repository scripts. The tiers form a ladder from the smallest solver-resistant pocket to the full upper-bound proof.

Tier	Artifact	Size	Resistance level	Recommended target
T1a	T1a JSONL	25 chunks	closed by HiGHS at 8h (dual = <code>inf</code>)	reference / regression test

Tier	Artifact	Size	Resistance level	Recommended target
T1b \ T1c	T1b-minus-T1c JSONL	18 chunks	LP-paradigm-resistant; closed by CDCL; all 18/18 emitted DRAT proofs verified by drat-trim	certified CDCL benchmark
T1c	T1c JSONL	2 chunks	resistant to CP-SAT, HiGHS LP-MIP, pure CDCL @ 12h, and windowed CDCL @ 4h	minimum-viable proof step
T2	T2 JSONL	6,071 chunks	survived 60-s CP-SAT broad pass with window bounds	full closure of the expansion residual
T3	deterministic generator	12,582,912 chunks	unprocessed remainder of the witness-split lattice	full upper-bound proof

The concrete filenames are `results/N212_K44_t1a25.jsonl`, `results/N212_K44_t1b_minus_t1c.jsonl`, `results/N212_K44_t1c2.jsonl`, and `results/N212_K44_window100k_unknowns.jsonl`.

Tier T1c is the campaign’s sharpest open problem: 2 chunks resistant to every tested solver paradigm under generous wall caps. A successful T1c closure either disproves $r_3(212) \leq 43$ (one **FEASIBLE**/SAT row suffices, after witness verification) or eliminates the audited four-paradigm-resistant residual, leaving the unit gap depending on the unprocessed remainder of T3.

Tier T2 is the canonical “close the campaign at the broad layer” target: closing all 6,071 chunks of the 100k expansion batch is necessary, though not sufficient, for a full upper-bound proof.

Tier T3 is the full upper-bound certificate: closing the entire depth-24 AP-pruned sweep. The instance generator (§2.2) emits the required chunks deterministically from the witness file and the OEIS b-file; the storage cost of the full sweep is dominated by output rather than input.

5.3. Minimum-viable proof requirements. A formal proof of $r_3(212) \leq 43$ from the released benchmark requires:

1. **Closure of tier T3.** Every chunk in the depth-24 AP-pruned sweep must return **INFEASIBLE** under a verified solver, or one chunk must return a **FEASIBLE** 44-element witness.
2. **Solver verification.** We recommend that any solver used for closure either produce machine-checkable proof certificates (DRAT, LRAT, or equivalent), or be independently reproduced under a second solver paradigm. This is a verification target rather than a condition already met by every row of the present campaign; §6.2 records the remaining certificate gaps.
3. **Constraint-set verification.** The 11,130 3-AP triple inequalities, the 22,154 window-cardinality inequalities, and the endpoint forcing must be checked against the formal definition of $r_3(N)$. The verifier `r3_verify.py` performs this check on any candidate witness; an analogous check for the constraint generation is included in the repository.

The minimum-viable result short of a full proof is closure of T1c under a solver paradigm that produces a usable bound certificate (e.g., LP dual values that improve under cuts, or a CDCL-style

unsatisfiability proof on a SAT encoding of the same constraints). This would establish that the hard pocket is not solver-architecture invariant after all, contradicting the conjecture of §4.5.

5.4. Approaches we could not test. The following techniques are plausibly stronger than CP-SAT and HiGHS on this pocket but were outside the scope of our campaign. We flag them as natural follow-on directions:

- **Fourier-analytic upper bounds.** Behrend-style constructions achieve $r_3(N) = N \cdot \exp(-c \cdot \sqrt{\log N})$ lower bounds; a matching Fourier-analytic upper bound for small N would give a certificate independent of combinatorial search. The Bloom–Sisask framework for $r_3(N) = O(N/(\log N)^{\{1+c\}})$ is asymptotic but the underlying density-increment machinery may yield finite- N bounds tighter than the OEIS window-cardinality family.
- **Multi-window partition bounds.** OEIS A003002 is used here as a single-window family $\text{sum}_{\{[a, a+L]\}} x_i \leq r_3(L)$. A partition-cardinality bound across multiple disjoint windows of varying length might cut the LP relaxation more aggressively than any single-window family.
- **SAT with proof logging.** A pure CNF encoding has already closed 18/20 T1b chunks using PySAT/CaDiCaL without OEIS window constraints. Follow-up proof-producing runs emitted DRAT proofs for all 18 CDCL-closed chunks, and `drat-trim` independently verified all 18/18. The research target is now T1c: try longer CDCL walls, different cardinality encodings, full-window encodings, or native proof-producing SAT solvers on chunks 40959 and 48895.
- **Lean/formal-proof-search benchmark.** The repository now includes a compact Lean 4 / Mathlib 4 formalization [dMU21, The20] under `lean/`: shared definitions (`R3Base.lean`), the verified 43-witness statement (`R3_212_Witness.lean`), and two AlphaProof-style T1c targets (`R3_T1c_40959.lean`, `R3_T1c_48895.lean`) with the expected answer `false`. These files are intended as a starting point for an A003002 / $r_3(N)$ entry in formal-conjecture repositories [Goo26]. Recent AlphaProof Nexus-style workflows have resolved Erdős problems of comparable complexity at low per-problem cost when a Lean target exists [TKS+26], so T1c is well-shaped for that paradigm.
- **Custom branch-and-bound.** A solver that exploits problem-specific symmetries and dominance relations — for instance, reflection symmetry after endpoint forcing, or domination of one window-bound by another at specific fixed-pin configurations — could prune branches that neither CP-SAT nor HiGHS recognize as redundant.
- **Symmetric Difference Encoding / Lasserre hierarchy at low degree.** A degree-4 or degree-6 SDP relaxation of the 3-AP-free constraint might separate the surviving T1 instances from feasibility where the LP relaxation cannot.

We do not advocate for any single direction. The benchmark release is intended to let specialists pick the technique closest to their toolkit.

5.5. Reproducibility and release. The campaign code, configuration, and result logs are released at the repository accompanying this preprint. Specifically:

- All Python sources implementing the architecture of §2 are present with module-level documentation.
- The SLURM scripts (`submit_*.sbatch`) and the driver `unity_handoff.sh` reproduce the full campaign on any SLURM-compatible cluster with OR-Tools and `highspy` installed.
- The verified 43-witness, the OEIS A003002 b-file used for window bounds, the broad-pass result logs, the recap residuals, the HiGHS attack logs, the T1a/T1b/T1c benchmark JSONLs, verified DRAT certificates, LRAT artifacts where emitted with `drat-trim -L`, the

Lean T1c proof-search targets, environment/version captures for the main solver stack, and the scripts needed to generate T3 are all included.

- Existing single-instance entry points (`r3_split_cpsat.py` and `r3_highs_attack.py`, and `r3_sat_attack.py`) can rerun any row of T1c, T1b, or T2 by chunk ID.

The large solver artifacts are archived on Zenodo at <https://doi.org/10.5281/zenodo.20463334>. This archive contains the CNF/DRAT proof artifacts for the 18 CDCL-closed T1b T1c chunks, the available LRAT outputs, verification summaries, SLURM logs, solver outputs, SHA256 manifests, and reconstruction instructions for split archives [Erg26].

We welcome correspondence on partial results: any solver run that closes a strict subset of T1 or T2 is a meaningful incremental contribution, even if the full upper bound remains open.

6. DISCUSSION

We close with three observations: what the architecture transfers to, where it breaks down, and what the campaign suggests about the broader proof-search problem class for r_3 upper bounds.

6.1. Reusability for adjacent N . The five-component architecture of §2 is parametric in N and K and depends on two external inputs: a verified $(K - 1)$ -element lower-bound witness for $r_3(N)$ and the OEIS A003002 b-file prefix for window-cardinality pruning at lengths $L \leq N - 1$. Both inputs are standard finite data at the current frontier: exact values through $N = 211$ are tabulated by Cariboni/OEIS A003002, and any new run also requires an explicit verified lower-bound witness at the target size. We expect the architecture to apply directly to $N \in \{213, \dots, 220\}$ once such witnesses are supplied, with three caveats:

1. The depth of the broad split ($d = 24$ for $N = 212$) is empirical and likely needs to grow with N . The number of feasible-after-pruning chunks at depth d scales roughly geometrically with the size of the witness, so the broad layer for larger N will emit more chunks, and per-chunk cost will also rise as the number of variables and 3-AP triples grows.
2. The $r_3(N)$ value drops slowly as N grows, so the equality $\sum x_i = K$ becomes harder to refute by simple cardinality arguments. We expect the UNK rate at the 60-s wall cap to grow with N .
3. The endpoint-forcing argument generalizes: if $r_3(N - 1) = K - 1$, then any K -element 3-AP-free subset of $[1, N]$ contains both endpoints. This is the only structural input from prior work that the architecture exploits; everything else is N -uniform.

Within the OEIS A003002 frontier, the architecture is therefore a drop-in tool for any incremental $r_3(N')$ upper-bound attempt, with the same caveat the present campaign documents: the hard pocket of §4 will likely have an analogue at larger N , possibly more severe.

6.2. Limitations. The campaign’s main limitation is the unresolved hard pocket itself. Even granting all five CP-SAT levers and the HiGHS substitution, the architecture cannot close $r_3(212) \leq 43$ without an additional idea. Section 5 frames this as an open problem; here we record three narrower limitations of the present implementation:

- **Machine-checkable certificates cover the CDCL-resolved T1b \ T1c chunks, but not the full campaign.** Follow-up proof-producing runs emitted DRAT proofs for all 18 CDCL-resolved T1b \ T1c chunks, and independent `drat-trim` verification confirmed all 18/18. The 25 HiGHS-closed T1a chunks and the ~93,929 broad-pass CP-SAT INFEASIBLE returns remain solver-attested rather than third-party-verified; closing this remaining gap would require either a SAT re-encoding of each chunk or a verified LP-duality-style certificate format we are not aware of in the open MIP ecosystem.

- **Compute coverage is incomplete.** The campaign processed 100,000 of the 12,582,912 AP-pruned depth-24 chunks at the broad layer, plus refinements on small samples. A full sweep would require orders of magnitude more worker-hours under the present parameters and was not justified by the available evidence, given the hard-pocket diagnosis.
- **Witness dependence is structural, not adversarial.** The broad split is anchored to one specific 43-witness. A different witness would induce a different chunk-ID space and possibly a different hard pocket. We did not test whether the hard pocket is witness-invariant; if it is not, alternative witnesses might bypass the T1c residual of §4.3.
- **The Lean benchmark is not yet upstreamed.** We did not find a public A003002 / $r_3(N)$ entry in `google-deepmind/formal-conjectures` as of the campaign date. The repository includes a Lean 4 formalization of the witness and T1c targets, but upstreaming and independent typechecking in that ecosystem remain follow-on work.

The third limitation is the most interesting follow-on: a witness-ensemble version of the broad split, in which the chunk space is the disjoint union of split-prefixes from multiple distinct 43-witnesses, would either reveal an easier alternative covering of the search space or confirm that the hard pocket is intrinsic to the constraint family rather than to one witness.

6.3. What the campaign suggests about r_3 upper-bound search. The dominant finding is now more precise than the original CP-SAT/HiGHS comparison. LP-paradigm methods — CP-SAT-style propagation on the bounded model and HiGHS LP-MIP — fail uniformly on the 20 T1b chunks with flat dual bounds, while CDCL closes 18/20 of them. The residual T1c set has size 2 and is the only audited pocket that remains resistant across every tested paradigm.

Two narrower suggestions follow from the lever inventory of §4.4. First, the window-cardinality family from OEIS A003002 is doing nearly all of the work that generic constraint propagation can do; the ~28-percentage-point reduction it produces is not matched by any other intervention at any cost. Second, the levers that *should* have moved the residual under standard MIP/CP intuition — variable reordering, pair-AND Tseitin propagators, walltime extension to the plateau — moved it by less than two percentage points each. The residual is not lying in wait for a smarter generic search.

The exception is the CDCL paradigm switch (§4.3), which closes 18/20 of T1b. We treat that as a qualitative change of proof system, not another lever inside the LP-relaxation family.

Combined, these suggest that the most productive next experiment is not another CP-SAT or MIP variant but either (i) a stronger SAT/proof-search attack on T1c, (ii) a tighter upper-bound family on $\sum x_i$ (Fourier-analytic, multi-window partition, or SDP-derived; §5.4), or (iii) a custom branch-and-bound that hard-codes the reflection symmetry and any per-instance dominance relations the generic solvers cannot infer. The benchmark release of §5 is built to support exactly this kind of follow-on.

We end with the working conjecture of §4.5: T1c corresponds to a low-dimensional pocket in the depth-24 assignment lattice along which the LP relaxation gap is tight and the integer infeasibility certificate, while present, is not learned by current CDCL encodings under the tested wall caps. A proof or refutation of this conjecture would be the cleanest mathematical outcome of the campaign that does not require closing the full unit gap directly.

APPENDIX A. COMPUTE ACCOUNTING

The retained logs provide measured solver wall time for the main diagnostics. Since CP-SAT tasks used 8 workers, the worker-hour estimates below multiply recorded solver-wall seconds by 8; for HiGHS, the analogous estimate is 8 threads per one-hour task. These are retained-log estimates, not exact SLURM accounting totals.

Layer	Worker-hours	Notes
Retained broad logs	~2,276	includes the 10k no-window run, the 10k window run, the 100k window run, and the [575, 675) A/B
Sample-100 refinement	~375	depth-40 plus one depth-48 tail
Sample-500 refinement	~1,978	depth-40, depth-48, depth-56, and depth-64 cleanup
Recap and worst-bucket A/Bs	~598	reorder, 300s walltime, 120s recap, 300s recap
HiGHS attack on 45 survivors	~360	1-h wall, 8 threads per task
Full T1 HiGHS long-wall audit	~2,002	45 T1 chunks, 8-h cap, job 58782313; estimated from retained seconds fields
T1b CDCL first run	~19.8	20 single-core tasks, retained JSON row time 71,216.85 seconds
T1b proof-producing rerun + T1c diagnostic	~120	proof emission + 4-cell T1c grid
drat-trim verification	~61	jobs 58952708, 59058393, and 59383874; all 18 CDCL-resolved T1b \ T1c chunks VERIFIED
Total retained logs	~7,850 worker-hours	excludes interactive debugging and logs not retained locally

The campaign fits within a single academic SLURM allocation on a CPU partition. SLURM job identifiers are retained in the table and in the released logs so that every worker-hour estimate can be traced to a specific recorded run.

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USE OF GENERATIVE AI

In the preparation of this work, the author used Anthropic Claude (Claude Sonnet 4.5 and Claude Opus 4.6, accessed via the Claude Code agent interface) and OpenAI Codex (GPT-5, accessed via the Codex agent interface) for drafting and revision assistance on the manuscript, repository scaffolding, and computational orchestration. The author reviewed and verified every claim and every line of code before submission, and takes full responsibility for the content. These tools did not generate mathematical claims, did not direct the campaign’s experimental design, and do not appear as authors.

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